

Evaluating Clinical and Genomic Predictors of Overall Survival in Lower-Grade Glioma

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Background: Lower-Grade Gliomas ¹

- Lower-grade gliomas (LGG) are primary brain tumors that often affect younger adults and have highly variable clinical outcomes.
- Compared with high-grade glioblastoma, LGG patients often have longer survival, but prognosis can differ substantially across patients.
- Some patients experience relatively slow disease progression, while others develop more aggressive disease and shorter survival.
- This variation makes LGG an important setting for survival analysis.

¹The Cancer Genome Atlas Research Network, 2015

Clinical and Molecular Prognostic Factors ²

- Traditional clinical prognostic factors include patient age, histologic grade, tumor type, and treatment history. However, molecular profiling has changed how LGG prognosis is understood.
- In particular, isocitrate dehydrogenase (IDH) mutation status and 1p/19q codeletion are now recognized as major molecular markers associated with LGG survival. Tumors with IDH mutations generally have better prognosis, while IDH-wildtype tumors are associated with poorer outcomes. In LGG, tumors with both IDH mutation and 1p/19q codeletion often have the most favorable prognosis.
- Therefore, modern LGG survival analysis should account for both clinical factors and molecular subtype.

²Eckel-Passow et al., 2015

- The Cancer Genome Atlas (TCGA) provides clinical and genomic data that allow survival outcomes to be studied alongside molecular tumor features.

³Louis et al., 2016; Taylor et al., 2018

Motivation: Genomic Instability³

- The Cancer Genome Atlas (TCGA) provides clinical and genomic data that allow survival outcomes to be studied alongside molecular tumor features.
- This project will focus on three genomic instability measures:
 - **Aneuploidy burden**: large-scale chromosomal copy-number alterations,
 - **Tumor mutation burden (TMB)**: overall number of mutations in the tumor genome, usually reported as mutations per megabase
 - **Microsatellite instability (MSI)**: instability in repetitive DNA regions.
- These features may capture additional tumor biology not fully explained by clinical factors or molecular subtype.

³Louis et al., 2016; Taylor et al., 2018

Research Question

Main Question

Do genomic instability measures independently predict overall survival in LGG after accounting for clinical factors and molecular subtype?

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- Specifically, this project evaluates whether aneuploidy burden, TMB, and MSI are associated with overall survival by comparing clinical predictors with molecular and genomic predictors using Cox proportional hazards (PH) models.
- The goal is to assess whether genomic instability provides additional prognostic information beyond established LGG risk factors.

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Dataset Overview

- The analysis uses TCGA lower-grade glioma clinical and molecular data.
- Patient-level clinical information and sample-level molecular information were merged using patient ID.
- The primary outcome is overall survival, defined by survival time in months. Death is the event of interest, and patients alive at last follow-up are treated as right-censored.

Variable Dictionary

Variable	Levels / Units	Category
time	Overall survival time, months	Outcome
event	0 = censored/alive, 1 = deceased	Outcome
age	Years	Clinical*
sex	Female, Male	Clinical*
grade	Grade II, Grade III	Clinical*
subtype	IDHmut-codel, IDHmut-non-codel, IDHwt, Unknown	Molecular†
tumor_type	Astrocytoma, Oligoastrocytoma, Oligodendroglioma	Clinical*
radiation_therapy	No, Yes, Unknown	Clinical*
race	White, Black, Asian, American Indian/Alaska Native, Unknown	Demographic*
ethnicity	Hispanic/Latino, Not Hispanic/Latino, Unknown	Demographic*
aneuploidy_score	Number of altered chromosome arms	Genomic†
tmb_nonsynonymous	Mutations per megabase, mut/Mb	Genomic†
tmb_quart	TMB quartile groups	Genomic†
msi_score_mantis	Unitless MSI score from MANTIS	Genomic†
msi_sensor_score	Unitless MSI score from MSIsensor	Genomic†
pfs_months	Progression-free survival time, months	Secondary outcome
pfs_status	0 = censored, 1 = progression	Secondary outcome

*Clinical/demographic factor; †Molecular/genomic factor.

- Overall survival status was converted into a binary event indicator: 0 = alive/censored and 1 = deceased.
- Missing categorical values were coded as “Unknown” when useful, so that patients were not unnecessarily excluded.
- Variables were converted to the appropriate types before analysis.

Final analytic cohort

Characteristic	Value
Patients	501
Deaths	125
Censored	376
Median age	41 years-old
Female / Male	220 / 281
Grade II / III	240 / 261

- Molecular subtype
 - IDHmut-codel: $n = 163$
 - IDHmut-non-codel: $n = 243$
 - IDHwt: $n = 92$
 - Unknown: $n = 3$
- Tumor type
 - Astrocytoma: $n = 193$
 - Oligoastrocytoma: $n = 127$
 - Oligodendroglioma: $n = 181$

Variable Distributions

- The median age was 41 years, with a broad range from adolescence to older adulthood.

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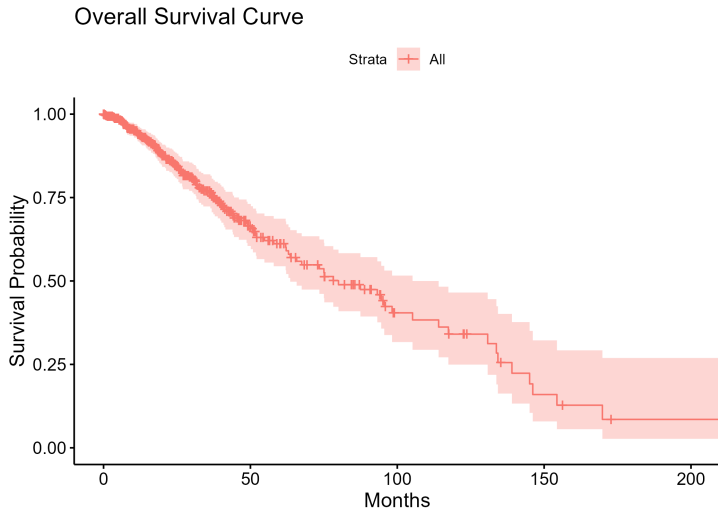
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- The cohort contains major molecular subtypes, allowing subtype to be evaluated as a central prognostic factor.

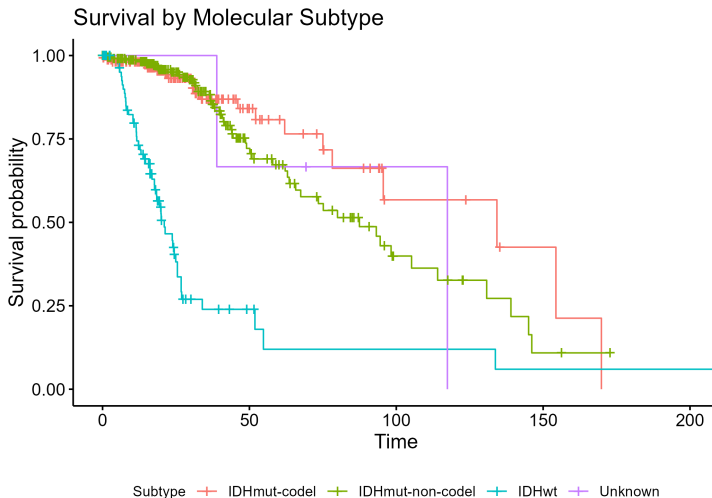
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- The cohort is predominantly White and non-Hispanic, so these variables were treated mainly descriptively rather than as major modeling targets.

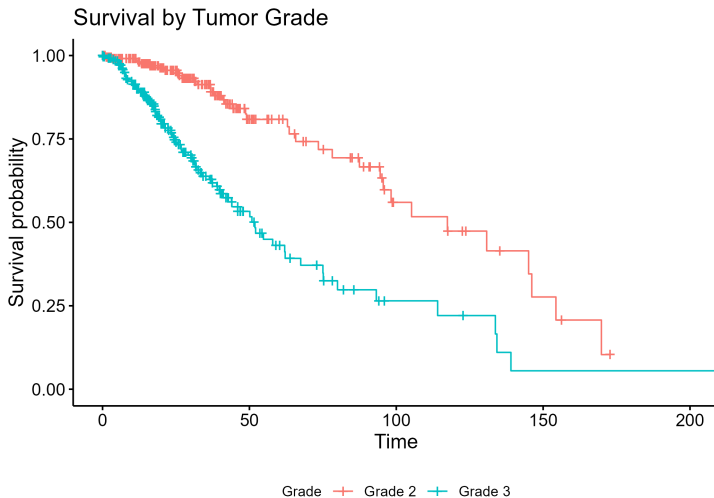
Overall Survival



KM Survival Curve of Molecular Subtype

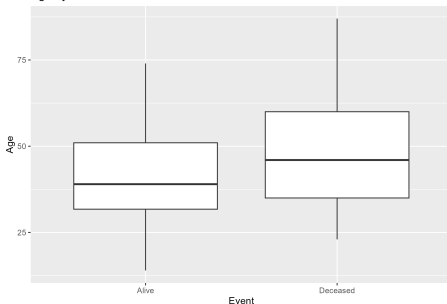


KM Survival Curve of Tumor Grade

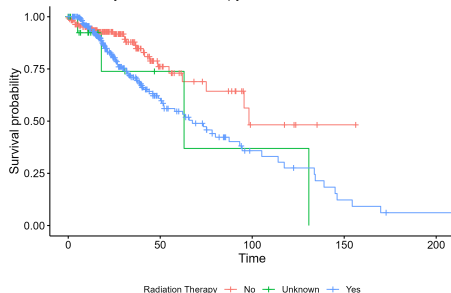


Clinical Predictors: Age and Treatment

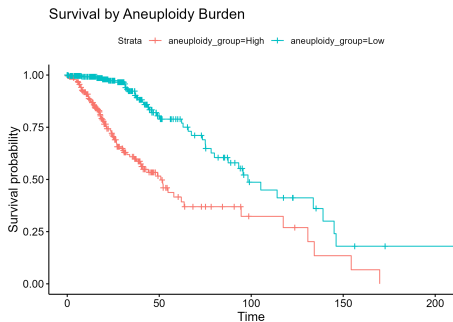
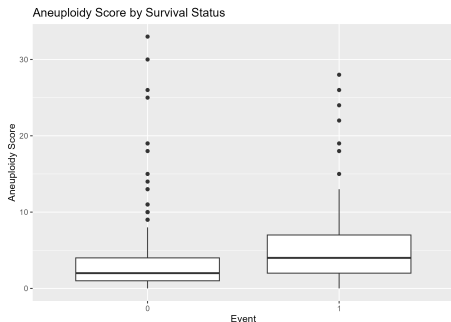
Age by Survival Status



Survival by Radiation Therapy



Molecular Predictors: Aneuploidy Burden



Molecular Predictors: TMB and Correlation Structure

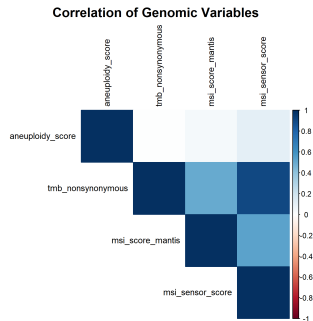
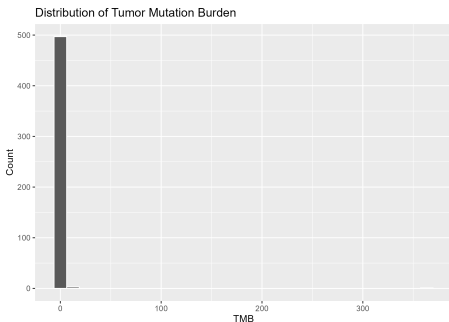


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Strategy Overview

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- 5 Compare candidate models using likelihood ratio tests, AIC, BIC, and concordance.

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- 5 Compare candidate models using likelihood ratio tests, AIC, BIC, and concordance.
- 6 Check PH assumptions and perform sensitivity analyses.

TMB Functional Form

- TMB was originally measured as a continuous molecular variable, but its distribution was extremely right-skewed.
- Martingale residual plots were used to assess whether TMB had an approximately linear relationship with the log hazard:
 - Original TMB was dominated by extreme outliers, making the pattern difficult to interpret.
 - $\log(\text{TMB} + 1)$ improved the scale, but the smoothed residual curve still did not show a clear linear trend.
 - To avoid forcing a linear effect, TMB was split into empirical quartiles.

Comparing TMB Specifications

Model	df	AIC	BIC
No TMB	12	1106.956	1140.896
Log TMB	13	1106.535	1143.030
TMB quartiles	15	1107.029	1149.540

- The AIC values were very similar across the three specifications.
- The log-TMB model had the lowest AIC, but the difference from the quartile model was small.
- BIC favored the simpler no-TMB model because it penalizes additional parameters more strongly.

TMB Quartile Groups

- Splitting TMB using empirical quartiles from the analytic cohort created four groups with roughly similar sample sizes:

TMB group	Range	n
Quartile 1	[0, 0.7]	142
Quartile 2	(0.7, 0.933]	115
Quartile 3	(0.933, 1.23]	120
Quartile 4	(1.23, 363]	124

- The first quartile was used as the reference group in the Cox model. Each higher quartile was compared against the lowest TMB group.
- This coding allows interpretation in terms of relative hazard for increasing TMB burden.

Clinical Model Comparison

- Model 1: age, sex, grade, radiation therapy, and molecular subtype.
- Model 2: age, sex, grade, radiation therapy, and tumor type.

Adding Molecular Predictors

- Candidate molecular predictors included aneuploidy score, TMB quartile, and MSI MANTIS score.
- Aneuploidy was considered because EDA suggested higher aneuploidy burden among patients with poorer survival.
- TMB was included as quartiles because of its extreme skewness and uncertain continuous functional form.
- MSI MANTIS score was selected as the MSI measure because MANTIS and MSIsensor were correlated, so including both could introduce redundancy.

Model Reduction and Final Model Choice

- A full clinical + molecular Cox model was first fitted.
- Variables that did not improve model fit or had limited interpretability were removed to obtain a reduced model.

Model Reduction and Final Model Choice

- A full clinical + molecular Cox model was first fitted.
- Variables that did not improve model fit or had limited interpretability were removed to obtain a reduced model.
- The reduced model retained age, grade, molecular subtype, TMB quartile, and MSI MANTIS score.
- Therefore, the final primary model was:

$$h(t | X) = h_0(t) \exp\{\beta_1 \text{Age} + \beta_2 \text{Grade} + \beta_3 \text{Subtype} + \beta_4 \text{TMB quartile} + \beta_5 \text{MSI}\}.$$

Model Comparison

Model	AIC	BIC	C-index
Clinical only	1115.1	1137.7	0.833
Full clinical + molecular	1104.3	1141.1	0.829
Reduced clinical + molecular	1098.0	1123.5	0.826

- The reduced clinical + molecular model had lower AIC and BIC than the clinical-only model.
- Concordance remained strong across models.
- The reduced model was chosen as the primary final model because it improved fit while remaining interpretable.

Time-varying MSI effect

- The PH diagnostic suggested that the MSI effect may vary over time.
- A sensitivity model included a time-dependent MSI term, $\text{MSI} \times \log(t)$.
- This model improved AIC from 1098.0 to 1094.8 and C-index from 0.826 to 0.834, suggesting improved fit, but was treated as a sensitivity analysis because interpretation is less straightforward.

Sensitivity Analyses: Model Structure

Subtype-stratified Cox model

- A stratified model using `strata(subtype)` was considered to allow each molecular subtype to have its own baseline hazard. This relaxes the PH assumption for subtype.
- However, stratifying by subtype means subtype itself no longer has a hazard ratio, so the main model retained subtype as a covariate for interpretability.

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Adding aneuploidy score

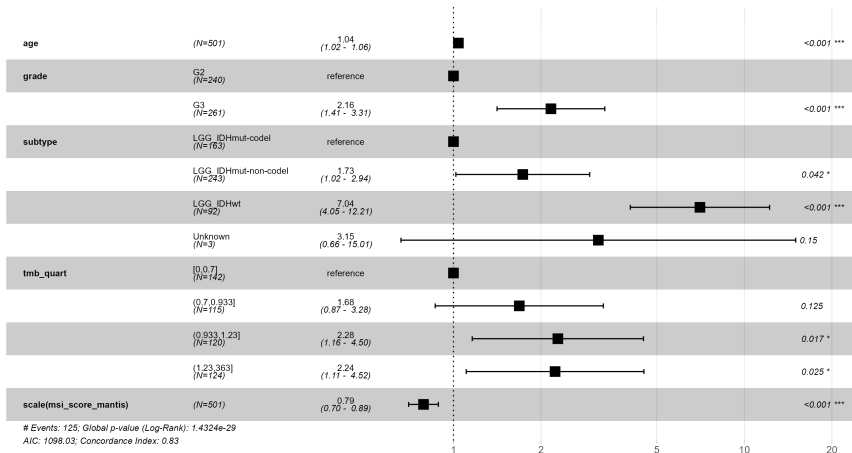
- Aneuploidy score was also evaluated as an additional molecular covariate.
- Although EDA suggested higher aneuploidy burden among poorer-outcome patients, it was not retained in the reduced final model after adjustment.
- Since the result suggests that its apparent effect may overlap with subtype, grade, TMB, MSI, or other clinical factors, it was not included in the final model.

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Final Cox Model: Main Results

Hazard Ratios from Final Cox Model

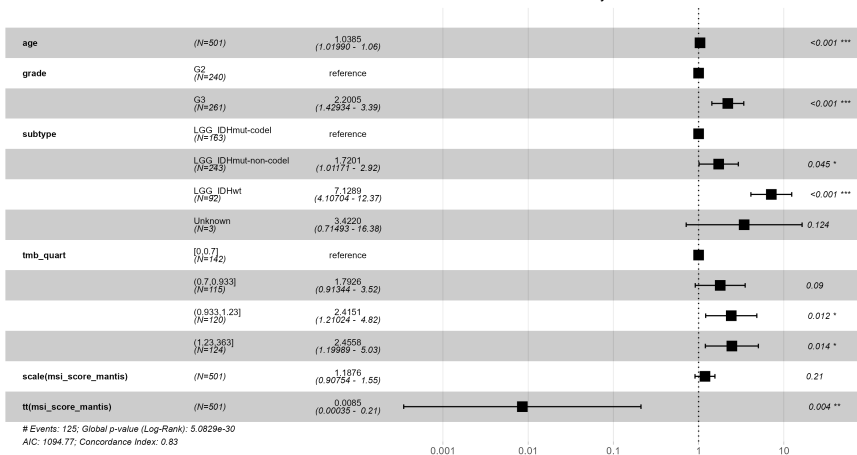


Final Cox Model: Interpretation

- The final primary model included age, grade, molecular subtype, TMB quartile, and standardized MSI MANTIS score.
- MSI MANTIS was standardized using `scale(msi_score_mantis)`, so its hazard ratio is interpreted per one-standard-deviation increase in MSI score.
- Older age, Grade III tumors, IDH-wildtype subtype, and higher TMB quartiles were associated with increased hazard of death.
- In the primary model, higher standardized MSI MANTIS score was associated with lower hazard, but this result should not be interpreted as a simple protective effect because the sensitivity analysis suggested that the MSI association may not be constant over time.

Time-Varying MSI Effect: Main Results

Hazard Ratios from Final Cox Model with Sensitivity



Time-Varying MSI Effect: Interpretation

- In this sensitivity model, the regular MSI term was no longer statistically significant ($HR = 1.19$, $p = 0.21$). However, the time-varying MSI term was statistically significant ($HR = 0.0085$, $p = 0.004$).
- Because this model is harder to interpret than the primary Cox model, it is reported as a sensitivity analysis rather than selecting it as the main final model.
- Importantly, the main conclusions for age, grade, molecular subtype, and TMB quartile remain similar.

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Key Findings

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Older age, Grade III disease, and IDH-wildtype subtype were associated with worse overall survival.

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Higher TMB quartiles were associated with increased hazard after adjustment for clinical factors and molecular subtype. MSI appeared protective in the primary Cox model, but sensitivity analysis suggested that the MSI association may vary over time.

Key Findings

Clinical predictors

Older age, Grade III disease, and IDH-wildtype subtype were associated with worse overall survival.

Molecular predictors

Higher TMB quartiles were associated with increased hazard after adjustment for clinical factors and molecular subtype. MSI appeared protective in the primary Cox model, but sensitivity analysis suggested that the MSI association may vary over time.

Main takeaway

Genomic instability measures, particularly TMB, added prognostic information beyond clinical factors and molecular subtype. However, established prognostic variables, especially tumor grade and IDH subtype, remained central to survival risk stratification.

Implications for Prognostic Prediction

- The final Cox model had strong discrimination, with a concordance index around 0.83.
- Age, grade, and molecular subtype provided a strong clinical prognostic foundation.
- Adding genomic instability measures, especially TMB quartiles, helped evaluate whether genomic features contributed additional prognostic information beyond clinical and subtype predictors. While MSI showed an association in the primary model, sensitivity analysis suggested that its prognostic effect may vary over time.
- However, this analysis should be interpreted as prognostic modeling and risk stratification, not as a fully validated clinical prediction tool.

- The results are consistent with the known importance of age, grade, and molecular subtype in lower-grade glioma prognosis.
- The strong effect of IDH-wildtype subtype supports the role of molecular classification in survival risk stratification.
- TMB showed evidence of prognostic relevance after adjustment, but its interpretation depends on how the highly skewed TMB distribution is modeled.
- MSI was significant in the primary model, but the time-varying sensitivity model suggested that its association with hazard may change over follow-up.

Limitations

- The analysis used observational data, so treatment-related associations should not be interpreted causally.
- Some categories were sparse, especially the unknown subtype group.
- Race and ethnicity analyses were limited because the cohort was predominantly White and non-Hispanic.
- PH diagnostics suggested possible non-proportional hazards for MSI.
- TMB was highly right-skewed, so conclusions may depend on how it is coded.
- The model was evaluated internally. External validation was not performed, so the model's generalizability remains uncertain.

- Validate the final Cox model in an independent LGG cohort to assess generalizability.
- Evaluate predictive accuracy using train-test splits or cross-validation.
- Explore flexible functional forms for continuous genomic predictors, such as splines for TMB and aneuploidy score.
- Further investigate time-varying effects, especially for MSI.
- Consider additional genomic features, treatment variables, and interaction terms, such as subtype-specific molecular effects.

Thank You!
Questions?

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